

1. Administrative & Study Identification

Full Study Title: A Phase 3 Study to Assess Efficacy Safety and Tolerability of Remibrutinib in Adult Patients With Moderate to Severe Hidradenitis Suppurativa **Official Regulatory Title:** A Randomized, Double-blind, Double-dummy, Placebo-controlled, Multicenter, Phase 3 Study Assessing the Efficacy, Safety, and Tolerability of 2 Doses of Remibrutinib Over a 68-week Treatment Period in Adult Patients With Moderate to Severe Hidradenitis Suppurativa. **Short Title/Acronym:** RECHARGE 1 **Protocol Number:** CLOU064J12301 **NCT Number:** NCT06799000 (Posting ID: 1088022346333617332) **Sponsor/Company:** Novartis **Investigational Product:** Remibrutinib (LOU064) **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To establish the efficacy, safety, and tolerability of remibrutinib (LOU064) Dose A and Dose B compared to placebo in participants with moderate to severe hidradenitis suppurativa (HS). **Secondary Objectives:** To evaluate further reductions in symptoms, including Abscesses and inflammatory nodules 50 (AN50) response, changes in International Hidradenitis Suppurativa Severity Score System (IHS4), and HiSCR75 response.

2.2 Background & Rationale To address the significant inflammatory burden in patients with moderate to severe Hidradenitis Suppurativa (HS). HS is a chronic, progressive skin disorder characterized by recurrent nodules, abscesses, draining tunnels, and scarring. Traditional treatments are often off-label or provide insufficient relief. Remibrutinib is a Bruton’s tyrosine kinase (BTK) inhibitor, providing a targeted approach that directly inhibits BTK—a key enzyme in the body's inflammatory signaling pathway. This mechanism aims to provide more effective and rapid relief from symptoms by addressing the underlying immunopathology of the condition.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind, Double-dummy **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** 555 participants
Number of Sites: Multicenter (Global) **Estimated Study Start:** 31-Jan-2025 **Estimated Primary Completion:** 31-Aug-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants ≥ 18 years of age at the time of signing of the informed consent. 2. Diagnosis of HS based on clinical history and physical examination for at least 6 months prior to the Baseline visit. 3. Participants with moderate to severe HS defined as: * A total of at least 5 AN count (abscesses and/or inflammatory nodules) AND * Inflammatory lesions should affect at least 2 distinct anatomic areas (e.g., left and right axillae).

4.2 Exclusion Criteria 1. Presence of more than 20 fistulae/tunnels (both draining and non-draining) in total at baseline. 2. Any active skin disease or conditions that may interfere with the assessment of HS. 3. Previous exposure to remibrutinib or other BTK inhibitors. 4. Use of other investigational drugs within 5 half-lives of enrollment, or within 30 days (for small molecules) prior to randomization, or until the pharmacodynamic effect has returned to baseline (for biologics), whichever is longer. 5. Significant bleeding risk or coagulation disorders. 6. History of gastrointestinal bleeding. 7. Requirement for anti-platelet (except for acetylsalicylic acid up to 100 mg/d or clopidogrel up to 75 mg/d) or anti-coagulant medication. 8. History or current hepatic disease. 9. Evidence of clinically significant cardiovascular, neurological, psychiatric, pulmonary, renal, hepatic, endocrine, metabolic, hematological disorders, gastrointestinal disease or immunodeficiency that would compromise the safety of the participant or preclude protocol adherence. 10. History of hypersensitivity to any of the study drug constituents. 11. Known or suspected infectious disease that is active, chronic or recurrent (e.g., tuberculosis, atypical mycobacterioses, listeriosis, aspergillosis, HIV). 12. History of live attenuated vaccine administration within 6 weeks prior to randomization or requirement to receive these vaccinations at any time while on study treatment. 13. Major surgery within 8 weeks prior to screening or planned surgery for the duration of the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Remibrutinib Dose A or Dose B vs. Placebo. **Route of Administration:** Oral (film-coated tablets). **Duration of Treatment:** 76 weeks total study duration (Screening up to 4 weeks; Treatment Period 1: 16 weeks, double-blind placebo-controlled; Treatment Period 2: 52 weeks, treatment with remibrutinib active doses; Safety Follow-Up: treatment-free follow-up for 4 weeks). Participants who prematurely discontinue

study treatment are encouraged to remain in the study; if not, they enter a 4-week Safety Follow-Up period.

5.2 Concomitant & Prohibited Therapy Permitted: Stable use of acetylsalicylic acid up to 100 mg/d or clopidogrel up to 75 mg/d. **Prohibited:** Use of other investigational drugs within 5 half-lives of enrollment, prior exposure to BTK inhibitors, requirement for anti-coagulant or prohibited anti-platelet medications, and live attenuated vaccines while on study treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to 4 weeks	Informed Consent, Medical History, Physical Exam, Baseline Lesion Count
Baseline	Day 0	Randomization, First Dose, Disease Assessment
Interim 1	Week 16	Primary Efficacy Assessment (HiSCR50), Safety Labs, Transition to Treatment Period 2
End of Study	Week 76	Final Vital Signs, Lesion Counts, IP Accountability, 4-Week Safety Follow-Up concluding the study

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants with Hidradenitis Suppurativa clinical response 50 (HiSCR50) at Week 16. HiSCR50 is defined as at least a 50% decrease in Abscess and Inflammatory Nodule (AN) count with no increase in the number of abscesses and in the number of draining tunnels/fistulae compared to baseline. **Measurement Method:** Clinical physical examination and standardized lesion counting.

7.2 Secondary Endpoints

- Proportion of participants with Abscesses and inflammatory nodules 50 (AN50) response at Week 16.
- Percentage change from baseline in International Hidradenitis Suppurativa Severity Score System (IHS4) at Week 16.
- Proportion of participants with HiSCR75 response at Week 16.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and physical check-ups throughout the 76-week study period. **Serious Adverse Events (SAEs):** Evaluated per standard protocol for Phase 3 clinical trials, reporting within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to review safety data (e.g., hepatic or cardiovascular signals) periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments; Safety Set for all participants receiving at least one dose of the investigational product.

Sample Size Justification: $N = 555$, adequately powered ($\alpha = 0.05$, $Power \geq 80\%$) to detect statistically significant differences in HiSCR50 response rates between the remibrutinib and placebo arms. **Handling of Missing Data:** Nonresponder imputation (NRI) standard methodology for binary categorical variables such as the HiSCR response endpoint.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits for protocol adherence, lesion count accuracy, and data source verification.

Audit Trail: Validated eCRF systems with comprehensive audit trails and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CLOU064J12301 **Registry Number:** NCT06799000 **Sponsor/Lead Organization:** Novartis **Study Title:** RECHARGE 1 **Official Regulatory Title:** A Randomized, Double-blind, Double-dummy, Placebo-controlled, Multicenter, Phase 3 Study Assessing the Efficacy, Safety, and Tolerability of 2 Doses of Remibrutinib Over a 68-week Treatment Period in Adult Patients With Moderate to Severe Hidradenitis Suppurativa.

12. Clinical Framework (Hidradenitis Suppurativa Specific)

Primary Condition: Hidradenitis Suppurativa (HS) **Diagnosis**

Threshold: Clinically confirmed HS for ≥ 6 months with ≥ 5 inflammatory nodules/abscesses affecting at least 2 distinct anatomic areas. **Medical Methodology:** Bruton's Tyrosine Kinase (BTK) Inhibition. **Optimization Target:** Achievement of HiSCR50 (reduction of inflammatory lesion burden).

13. Study Procedures & Methodology

Management Pathway: Standard clinical assessment of HS severity, lesion counting, and regular physical examinations. **Education**

Protocol (HCP): Site initiation training on accurate HS lesion identification (nodules, abscesses, draining fistulae). **Education**

Protocol (Patient): Education on reporting symptoms, daily oral

medication adherence, and recognizing potential side effects.
Quality Audit Mechanism: Centralized data review for accuracy of HiSCR50 endpoint calculation.

14. Observation & Follow-Up Schedule

Total Duration: 76 weeks (inclusive of up to 4 weeks screening, 68 weeks treatment across two periods, and 4 weeks safety follow-up). **Onsite Visit Cadence:** Regular intervals across Screening, Baseline, Week 16, Week 68, and Week 76. **Remote Monitoring Frequency:** As indicated by site protocols and risk-based monitoring frameworks. **Checklist Review Interval:** Routine continuous assessment of concomitant medications and adverse events.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					